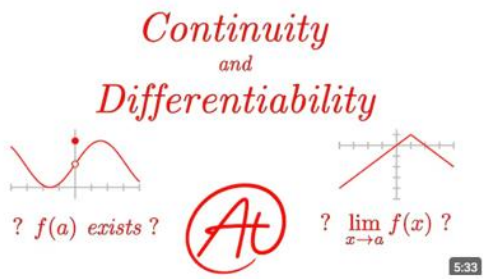
MATH I – Week 5

Extra (Multimedia) Material

1. The link to YouTube presentation on Continuity, which was presented in class (Week 5 Lecture) appears below. The presentation presents “continuity” (covered this week) and “differentiability” (which will be covered next week).

For now, you may watch the first half of the presentation (on continuity).



Continuity and Differentiability EXPLAINED with Examples
288K • 4y ago
Ace Tutors
Learn the meaning behind **Continuity** and Differentiability. This video explains what **Continuity** is and the 3 rules for a **function**
4K College

Link: https://www.youtube.com/watch?v=nRv1lPqrx3k

2. This week, Wikipedia’s lemma on Weierstrass was also presented:

Karl Weierstrass

71 languages

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From Wikipedia, the free encyclopedia

Karl Theodor Wilhelm Weierstrass (/ˈvaɪər ˌstrɑːs, -ˌʃtrɑːs/;[1] German: *Weierstraß* [ˈvaɪɐʃtʁaːs];[2] 31 October 1815 – 19 February 1897) was a German mathematician often cited as the "father of modern analysis".[3] Despite leaving university without a degree, he studied mathematics and trained as a school teacher, eventually teaching mathematics, physics, botany and gymnastics.[4] He later received an honorary doctorate and became professor of mathematics in Berlin.

Among many other contributions, Weierstrass formalized the definition of the continuity of a function and complex analysis, proved the intermediate value theorem and the Bolzano–Weierstrass theorem, and used the latter to study the properties of continuous functions on closed bounded intervals.



If you are interested to learn more, go to: https://en.wikipedia.org/wiki/Karl_Weierstrass

Note: The Weierstrass theorem presented in class is not the same as the Bolzano-Weierstrass theorem (we are not going to cover this B-W theorem).

Is the Bolzano-Weierstrass theorem the same as the Weierstrass theorem?

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AI Overview

No, the Bolzano-Weierstrass theorem is not the same as the Extreme Value Theorem (often called the Weierstrass extreme value theorem), though both are foundational results in mathematical analysis named in part after Karl Weierstrass. Wikipedia +1